

Financial Alpha & Risk Research Lab

This document provides a detailed, implementation-oriented description of the Financial Alpha & Risk Research Lab. The Lab is a Motia-orchestrated quant research environment that unifies multi-asset data ingestion, factor modeling, strategy research, portfolio construction, and risk analytics into a reproducible, governed platform for professional investors.

1. Problem & Vision

Institutional investors operate on fragmented stacks: separate scripts for data pulls, ad hoc spreadsheets for factors, one-off backtest frameworks, and limited traceability between research and production. This leads to:

- Inconsistent results between researchers and environments.
- Limited ability to reproduce historical decisions and strategies.
- Slow iteration cycles when testing new datasets or models.
- Weak integration between research outputs and risk/oversight functions.

The Financial Alpha & Risk Research Lab addresses this by providing a single, structured environment where data, factors, strategies, backtests, optimizations, and risk metrics are defined as code, orchestrated as workflows, and fully tracked. It is designed to plug into existing OMS/PMS/RMS setups without forcing a rip-and-replace.

2. Core Outcomes

The Lab is built to deliver the following outcomes:

- 1) Clean, Centralized Research Data
 - Normalized, survivorship-bias-aware, corporate-action-adjusted datasets.
 - Consistent identifiers across venues and data vendors.
- 2) Structured Factor & Signal Engineering
 - Shared library of technical, fundamental, macro, and alternative signals.
 - Versioned definitions so research is comparable and auditable.
- 3) Robust Backtesting & Simulation
 - Realistic models of fills, slippage, spreads, borrow, and latency.
 - Scenario and regime testing as first-class citizens.
- 4) Integrated Portfolio & Risk Analytics
 - Optimization pipelines that align with real desk constraints.
 - Risk metrics consumable by both PMs and independent risk teams.
- 5) Automated Strategy Search with Guardrails
 - Optuna and evolutionary loops exploring parameter and structural choices.
 - Embedded robustness checks to reduce overfitting.
- 6) Transparent, Governed Research
 - Every result traceable to data versions, configs, and code revisions.

3. Architecture Overview

The Lab is centered on Motia as the workflow and event orchestration layer. Around it:

- Data tier:
 - PostgreSQL for metadata (strategies, runs, configs, permissions).
 - DuckDB or ClickHouse for large time series and factor tables.
 - Object storage for raw vendor dumps, alternative datasets, and artifacts.

- Compute tier:
 - PyTorch for ML models (forecasting, classification, embeddings).
 - vectorbt / Backtrader for backtesting engines.
 - PyPortfolioOpt and cvxpy for portfolio optimization.
 - River + ADWIN for online learning and drift detection.
- Intelligence & retrieval:
 - OpenSearch for logs, metrics, and searching documents (filings, transcripts).
 - Qdrant for dense retrieval over textual research content.
 - Neo4j for issuer and relationship graphs (issuers, sectors, holdings, entities).
- Governance & security:
 - Vault for secrets and credentials.
 - OPA for policy-based access controls by desk, asset class, and role.
- Observability & lineage:
 - MLflow for experiments, models, and parameters.
 - Langfuse for tracing LLM/RAG flows in the research assistant components.
 - Prometheus and Grafana for infra and pipeline metrics.

4. Key Workflows

4.1 Market & Fundamentals Ingestion

- Scheduled Motia flows pull prices, volumes, fundamentals, macro series, and alternative data.
- All series are adjusted, normalized, and keyed on stable identifiers.
- Each run emits events (e.g., prices.updated) to trigger factor recomputations.

4.2 Factor & Signal Computation

- Factor definitions are implemented as composable functions with metadata.
- Pipelines compute cross-sectional and time-series factors on schedules or on demand.
- Outputs are stored in columnar form to support fast slicing by date, universe, and factor.

4.3 Backtesting & Simulation

- Strategies are defined as config + code: universe, signals, rules, execution model.
- Backtests compute returns, risk metrics, and diagnostics (e.g., exposures, turnover).
- All runs are logged with configs and code hashes for reproducibility and review.

4.4 Portfolio Construction & Risk

- Inputs: expected returns (from models or heuristics), covariance estimates, constraints.
- The engine supports mean-variance, risk parity, and custom objective functions.
- Risk analytics include VaR/CVaR, drawdown statistics, factor exposure breakdowns, and scenario shocks (e.g., rate shifts, volatility spikes, spread widening).

4.5 Strategy Evolution & Robustness

- Optuna-driven HPO sweeps hyperparameters and signal weights.
- OpenEvolve evolves rule sets, entry/exit thresholds, and portfolio construction logic.
- Each candidate strategy is evaluated on:
 - In-sample and out-of-sample performance.
 - Multiple regimes (bull, bear, sideways).
 - Tail risk and capacity constraints.
- Only candidates passing robustness gates are promoted.

4.6 Research RAG & Narrative Layer

- A retrieval layer lets users query: "Why did this strategy underperform in 2022?"
- The system combines PnL breakdowns, factor moves, filings, and macro narratives.
- Outputs are designed to be reviewable by IC and risk committees.

5. Security & Governance

The Lab is designed for sensitive environments:

- Authentication delegated to enterprise identity (OIDC/SAML).
- OPA policies define who can access which datasets, universes, and strategies.
- Vault stores credentials for brokers, data providers, and infrastructure.
- Network-level controls isolate the platform in hardened environments.
- Full audit trails exist for data ingestions, code changes, model runs, and approvals.

6. Implementation & Extensibility

Phase 1: Core ingestion, standardized factor engine, basic backtesting, MLflow integration.

Phase 2: Full portfolio and risk engine plus research RAG search across filings and notes.

Phase 3: Strategy evolution framework with Optuna/OpenEvolve and robustness gates.

Phase 4: Integration with clients' OMS/PMS/RMS systems and custom reporting surfaces.

The architecture deliberately separates orchestration (Motia), analytics libraries, and data storage, allowing different clients to adopt only the layers they need while preserving the same reproducible research model.